

Unit 1 Progress Check: FRQ

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to **K**.

- (a) In terms of atomic structure, explain why the atomic radius of **K** is larger than that of **Na**.



Please respond on separate paper, following directions from your teacher.

- (b) In terms of atomic structure, explain why the first-ionization energy **K** is less than that of **Ca**.



Please respond on separate paper, following directions from your teacher.

- (c) **K** forms the compound **K₂O**, which is an ionic compound that is brittle. Identify another element, **M**, that is likely to form a brittle, ionic compound with the formula **M₂O**. Justify your answer in terms of periodic trends.



Please respond on separate paper, following directions from your teacher.

- (d) The compound **K₂O₂** also exists. A chemist can determine the mass of **K** in a sample of known mass that consists of either pure **K₂O** or pure **K₂O₂**. From this information, can the chemist answer the question of which compound is in the sample? Indicate yes or no, and explain.



Please respond on separate paper, following directions from your teacher.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response indicates one of the following:

- ☐ that the valence electrons of **K** are in a higher energy shell than those of **Na** are
- ☐ that the outer orbitals have a higher principal quantum number in **K** than in **Na**
- ☐ that the valence electrons of **K** are less attracted to the nucleus than those of **Na**

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response indicates that the first ionization energy of **K** is less than that of **Ca** because the electron removed from **K** is less strongly attracted to (and thus more easily removed from) the **K** nucleus, which has fewer protons than **Ca**.

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response meets both of the criteria below:

- ☐ **Li, Na, Rb, Cs, or Fr**.
- ☐ the element chosen is in the same group/family/column of the periodic table as **K** OR that an atom of the element the student chose has the same number of valence electrons as a **K** atom.



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Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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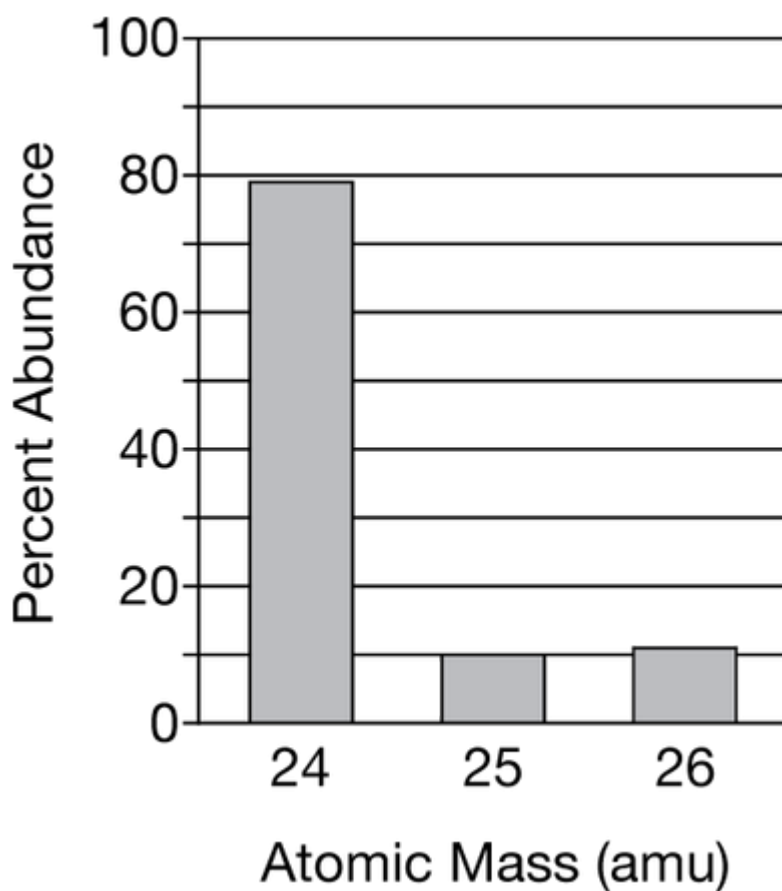
The response meets both of the criteria below:

- ☐ the chemist can determine the identity of the compound.
 - ☐ if the mass of **K** in a sample of the compound is determined, then the percent mass of **K** in the compound can be calculated and the compound identified because the percent mass is different (or known) for the two compounds.
2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A sample of a pure element is analyzed using a mass spectrometer. The results are shown below.



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(a) How many different isotopes of the element were in the sample?



Please respond on separate paper, following directions from your teacher.

(b) Describe how to use the information from the mass spectrum to determine the average atomic mass of the element.



Please respond on separate paper, following directions from your teacher.

(c) Identify the element.



Please respond on separate paper, following directions from your teacher.



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- (d) Write the ground-state electron configuration of an atom of the element that you identified in part (c).



Please respond on separate paper, following directions from your teacher.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response indicates that there are three isotopes represented in the mass spectrum.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response indicates that the weighted average must be calculated by finding the sum of three terms, each of which is the product of the proportional abundance of each isotope and the mass of each isotope

(e. g. , $(0.79)(24) + (0.10)(25) + (0.11)(26)$, or $(0.80)(24) + (0.10)(25) + (0.10)(26)$)

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response indicates that the element is **Mg**.

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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The response indicates only one of the following:

- ☐ the configuration $1s^2 2s^2 2p^6 3s^2$.
- ☐ the correct configuration for the element identified in part (c).